

Ch32: 3, 45, 49

Basic score 4` +problem 49 6`

3. (a) Application of equation $\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$ along the circle inside the plate, we have:

$$B(2\pi r) = \epsilon_0 \mu_0 \frac{d}{dt}(E\pi r^2) = \epsilon_0 \mu_0 \pi r^2 \frac{dE}{dt} \Rightarrow B = \frac{1}{2} \epsilon_0 \mu_0 r \frac{dE}{dt}.$$

For $r = 0.0200$ m, this gives

$$\begin{aligned} B &= \frac{1}{2} \epsilon_0 \mu_0 r \frac{dE}{dt} = \frac{1}{2} \epsilon_0 \mu_0 (0.020\text{m})(4.5 \times 10^{-3} \text{V/m}) \\ &= 5.01 \times 10^{-22} \text{ T} \end{aligned}$$

(b) With $r > R$, the expression above must be replaced by

$$B(2\pi r) = \epsilon_0 \mu_0 \pi R^2 (0.00450 \text{ V/m} \cdot \text{s}).$$

Substituting $r = 0.050$ m and $R = 0.040$ m, we obtain $B = 8.01 \times 10^{-22}$ T.

$$\text{c) } \begin{cases} B = \frac{\epsilon_0 \mu_0}{2} \frac{dE}{dt} r & (r < R), \quad r \uparrow, B \uparrow \\ B = \frac{\epsilon_0 \mu_0 R^2}{2r} \frac{dE}{dt} & (r \geq R), \quad r \uparrow, B \downarrow \end{cases}$$

So at $r = R$, the field magnitude is maximum.

45. 电流按照 100A 来计算

(a) Using Eq. 26-11, we find

$$E = \rho J = \frac{\rho i}{A} = \frac{(1.62 \times 10^{-8} \Omega \cdot \text{m})(100 \text{ A})}{5.00 \times 10^{-6} \text{ m}^2} = 0.324 \text{ V/m}.$$

(b) The displacement current is

$$\begin{aligned} i_d &= \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 A \frac{dE}{dt} = \epsilon_0 A \frac{d}{dt} \left(\frac{\rho i}{A} \right) = \epsilon_0 \rho \frac{di}{dt} = (8.85 \times 10^{-12} \text{ F/m})(1.62 \times 10^{-8} \Omega)(2000 \text{ A/s}) \\ &= 2.87 \times 10^{-16} \text{ A}. \end{aligned}$$

$$\text{(c) The ratio of fields is } \frac{B(\text{due to } i_d)}{B(\text{due to } i)} = \frac{\mu_0 i_d / 2\pi r}{\mu_0 i / 2\pi r} = \frac{i_d}{i} = \frac{2.87 \times 10^{-16} \text{ A}}{100 \text{ A}} = 2.87 \times 10^{-18}.$$

电流按照 54A 来计算

(a) Using Eq. 26-11, we find

$$E = \rho J = \frac{\rho i}{A} = \frac{(1.62 \times 10^{-8} \Omega \cdot \text{m})(54 \text{ A})}{5.00 \times 10^{-6} \text{ m}^2} = 0.175 \text{ V/m}.$$

(b) The displacement current is

$$\begin{aligned} i_d &= \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 A \frac{dE}{dt} = \epsilon_0 A \frac{d}{dt} \left(\frac{\rho i}{A} \right) = \epsilon_0 \rho \frac{di}{dt} = (8.85 \times 10^{-12} \text{ F/m})(1.62 \times 10^{-8} \Omega)(2000 \text{ A/s}) \\ &= 2.87 \times 10^{-16} \text{ A}. \end{aligned}$$

(c) The ratio of fields is $\frac{B(\text{due to } i_d)}{B(\text{due to } i)} = \frac{\mu_0 i_d / 2\pi r}{\mu_0 i / 2\pi r} = \frac{i_d}{i} = \frac{2.87 \times 10^{-16} \text{ A}}{54.0 \text{ A}} = 5.32 \times 10^{-18}.$

49. (a) At any instant the displacement current i_d in the gap between the plates equals the conduction current i in the wires. Thus $i_{\max} = i_{d \max} = 3.80 \mu\text{A}$. **0.5 point**

(b) Since $i_d = \epsilon_0 (d\Phi_E/dt)$,

$$\left(\frac{d\Phi_E}{dt} \right)_{\max} = \frac{i_{d \max}}{\epsilon_0} = \frac{3.80 \times 10^{-6} \text{ A}}{8.85 \times 10^{-12} \text{ F/m}} = 4.29 \times 10^5 \text{ V} \cdot \text{m/s}. \quad \text{1 point}$$

(c) Let the area plate be A and the plate separation be d . The displacement current is

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 \frac{d}{dt} (AE) = \epsilon_0 A \frac{d}{dt} \left(\frac{V}{d} \right) = \frac{\epsilon_0 A}{d} \left(\frac{dV}{dt} \right). \quad \text{1 point}$$

Now the potential difference across the capacitor is the same in magnitude as the emf of the generator, so $V = \epsilon_m \sin \omega t$ and $dV/dt = \omega \epsilon_m \cos \omega t$. Thus,

$i_d = (\epsilon_0 A \omega \epsilon_m / d) \cos \omega t$ and $i_{d \max} = \epsilon_0 A \omega \epsilon_m / d$. This means

$$d = \frac{\epsilon_0 A \omega \epsilon_m}{i_{d \max}} = \frac{(8.85 \times 10^{-12} \text{ F/m}) \pi (0.180 \text{ m})^2 (130 \text{ rad/s}) (220 \text{ V})}{3.80 \times 10^{-6} \text{ A}} = 6.78 \times 10^{-3} \text{ m},$$

1 point

(d) We use the Ampere-Maxwell law in the form $\oint \vec{B} \cdot d\vec{s} = \mu_0 I_d$, where the path of integration is a circle of radius r between the plates and parallel to them. I_d is the displacement current through the area bounded by the path of integration. Since the displacement current density is uniform between the plates, $I_d = (r^2/R^2)i_d$, where i_d is the total displacement current between the plates and R is the plate radius. The field lines are circles centered on the axis of the plates, so $\vec{B}$ is parallel to $d\vec{s}$. The field has constant magnitude around the circular path, so $\oint \vec{B} \cdot d\vec{s} = 2\pi rB$. Thus,

$$2\pi rB = \mu_0 \left(\frac{r^2}{R^2} \right) i_d \Rightarrow B = \frac{\mu_0 i_d r}{2\pi R^2}. \quad \text{2 point}$$

The maximum magnetic field is given by

$$B_{\max} = \frac{\mu_0 i_{d\max} r}{2\pi R^2} = \frac{(4\pi \times 10^{-7} \text{ T} \cdot \text{m/A})(3.80 \times 10^{-6} \text{ A})(0.110 \text{ m})}{2\pi (0.180 \text{ m})^2} = 2.58 \times 10^{-12} \text{ T}. \quad \text{0.5 point}$$